Code No: 138AC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, September - 2022
ADVANCED ALGORITHMS
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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JNTUH Used Papers

1.a) What is optimal binary search tree? Give dynamic programming formulation for the same.
b) How Partition is done in Quick Sort? Explain with an example. [7+8]

2.a) Discuss in detail about randomized algorithms.
b) Write an algorithm/pseudocode to sort elements using Heap sort technique. Illustrate the working of Heap sort algorithm on the following input: 35,15,0,1,60. [7+8]

3.a) How Dynamic Programming is used to solve Knapsack Problem? Explain.
b) Define the Single Source Shortest Path Problem and explain any one algorithm. [7+8]

4.a) Write Huffman code algorithm and derive its complexity.
b) Illustrate about minimum spanning trees with suitable examples. [7+8]

5.a) What is Strassen's matrix multiplication? Explain with example. Why is Strassen matrix multiplication considered better than simple matrix multiplication? Discuss.
b) Discuss in detail about Zero-one principle with example. [7+8]

6.a) Compute inverse of $A = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & -2 & 0 \\ 0 & 0 & -1 & 2 \end{pmatrix}$.
b) Write a detailed note on sorting networks. [7+8]

7.a) What is naive string matching algorithm? Write the complexity of naïve string matching.
b) Discuss the purpose of Knuth Morris Pratt algorithm? What problem does it solve? Explain. [7+8]

8.a) Explain Traveling Salesman problem with suitable example.
b) List and discuss various approximation algorithms in brief. [7+8]

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